

from last class:

$L: A \rightarrow B$ A, B must be vector spaces
(satisfy ①-⑥)

Ex: $V = \{\text{arrows on a piece of paper}\}$ i.e.

$$F = \mathbb{R}$$

$$x = \nearrow$$

$$y = \searrow$$

$$x+y = \nearrow + \searrow = \text{triangle}$$

$$y+x = \searrow + \nearrow = \text{triangle}$$

if addition is commutative then
 $y+x = x+y$

$$\text{parallelogram rule for } x+y$$

$$2.5x = \nearrow$$

if $|\vec{x}| = 1$ then $|2.5\vec{x}| = 2.5$

Do we have a 0 vector? $0 + \vec{x} = \vec{x}$
yes!

Ex: if F is any field, then lists of elements of length n form a vector space, w/ scalars from F

$$\hookrightarrow \mathbb{R}^2 = (_, _) \quad (a, b) + (c, d) = (a+c, b+d)$$

$$\mathbb{C}^4 \ni (0, i, 1+i, 17i) \quad i(1, i, 16+2i) = (i, -1, 16i-2)$$

Def: A set F is a field if:

- there is an operation $+$: $F \times F \rightarrow F$
- commutative
- associative
- has an identity "0"
- has inverses

- has an operation $\times$: $F \times F \rightarrow F$

- commutative
- associative
- has an identity "1"
- everything except 0 has a multiplicative inverse
- distributive laws

Def! $\mathbb{Q} := \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0 \right\}$

Def! A vector subspace of a vector space V is a subset of V which is also a vector space

EX! $V = \left\{ \begin{array}{l} \text{arrows in 2D} \\ \text{scalars} = \mathbb{R} \end{array} \right\}$ $W = \left\{ \begin{array}{l} \text{horizontal arrows} \\ \text{scalars} = \mathbb{R} \end{array} \right\}$

addition and multiplication are inherited from V

PF! $\vdash$ ident: $\longrightarrow + \cdot \stackrel{?}{=} \longrightarrow \text{yes!}$

$$\longrightarrow + \longleftarrow = \longrightarrow$$

$$\longrightarrow + \longleftarrow = \cdot$$

Theorem! if V is a vector space and $W \subset V$ then W is a vector subspace if it is closed under addition and scalar multiplication

PF! all the defining features of a vector space except closure are proved identically as for V

if $a \in F$ and $x \in W$ then $ax \in W$
if $x, y \in W$ then $x+y \in W$

EX! arrows in 2D which have length 1 are NOT a subspace

ex: $V = \{\text{differentiable functions}\}$

$W = \{\text{Polynomials of degree } n\}$